contributing foundational technology for the Anglo-French Future Combat Air System (FCAS) program.

Design and technical overview

The Taranis UCAV features a flying-wing configuration with smooth, blended surfaces designed to minimize radar cross-section (RCS) and infrared signature. Its airframe, constructed primarily from carbon-fiber composites and radar-absorbent materials (RAM), enhances stealth characteristics while maintaining structural integrity at high subsonic speeds.

With a wingspan of approximately 9.9 meters (32 feet) and a length of 11.3 meters (37 feet), the aircraft achieves a balance between agility, endurance, and payload capacity. The propulsion system is a Rolls-Royce Adour Mk 951 turbofan engine, adapted for stealth operation through modified air intakes and exhausts that reduce radar and heat emissions.

Taranis is capable of autonomous taxiing, take-off, flight, navigation, target detection, and landing, guided by advanced onboard systems rather than continuous remote control. This autonomy allows it to execute missions with minimal operator intervention, a key step toward future combat systems' human–machine teaming (HMT) capabilities.

Technological innovations and systems

1. *Stealth and signature reduction*: Taranis employs low-observable (LO) technology, including faceted airframe geometry, internal weapon bays, and advanced radar-absorbent coatings. These features allow the UCAV to penetrate defended airspace undetected.
2. *Autonomous mission control*: A major breakthrough in the Taranis program was its ability to plan, execute, and adapt missions autonomously. The onboard AI-driven mission management system